MIL-C-60632C (MU)
3 March 1970
SUPERSEDING
MIL-C-60632B (MU)
3 March 1969

MILITARY SPECIFICATION

CARTRIDGE, 40MM: XM576E1 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers Cartridge, 40MM, XM576E1, Loading Assembling and Packing.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-116	- Preservation, Methods of
MIL-P-20444	- Primer, Percussion, M42 Loading, Assembling and Packing
MIL-P-46610	- Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D10542398	- Cartridge, 40MM, XM576E1
C8835104	- Wooden Packing Box
C8835105	- Carton
FA-32821	- Marking Instructions for Wood Packing Box
FA-32822	- Marking Instructions for Fiber Packing Box and Bag
IEL-10542398	- Index of Inspection Equipment Lists for Cartridge, 40MM, XM576E1, Loading, Assembling and Packing

PUBLICATIONS

TECP 700-700 Interim Pamphlet 20-20 - U. S. Army Test and Evaluation Command Material Test Procedures Hand and Shoulder Weapons

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 Plastic defects.- All components and assemblies shall be free from blisters, bubbles, cracks, crazing, mottle, porosity, cold welds or any other defects.

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3.3 Metal Defects.- All components and assemblies shall be free from cracks, splits, cold shuts, inclusions, porosity or any other defects.

3.4 Loading

3.4.1 Propellant.- Propellant shall comply with the requirements of the applicable drawing.

3.4.2 Propellant weight.- The propellant weight shall be established and controlled to meet the recoil momentum. The established weight of the propellant charge shall not vary by more than ± 1/2 grain.

3.4.3 Propellant leakage.- Loaded base plugs shall show no evidence of propellant leakage.

3.4.4 Primer.- Primer shall comply with the requirements of the applicable specification.

3.4.4.1 Primer Functioning.- The primer shall function without evidence of squibbing or hangfire when tested as specified in 4.4.8. Paragraph 6.6 of MIL-P-20444B, Amendment 3, contains definitions of functioning failures.

3.4.5 Foreign matter.- Prior to loading, the cartridge case and base plug shall be free of dirt, contamination, oil, water or other defects detrimental in ballistics.

3.4.6 Examination.- After loading, the base plug shall be free of propellant grains on seating surfaces.

3.4.7 Primer misfire test (after insertion of base plug into cartridge case).- Base plug-cartridge case assemblies shall be fired and examined for the following characteristics.

Base Plug set-back in excess of 0.010 above flush with case

Primer Misfire

No Propellant

Primer leak as evidenced by smoke deposit or discoloration of base plug or cartridge case base.

Blown Primer as evidenced by perforation of the cup and extrusion of the primer cup around firing pin causing difficulty in opening weapon.

Loose Primer

Primer fallout of pocket

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3.5 Waterproof test.- The cartridges shall not fail to fire or to hand extract after being immersed in a 6 inch head of water for 12 hours.

3.6 Maximum profile.- The maximum profile of the cartridge shall comply with the applicable drawing.

3.7 Functional

3.7.1 Temperature condition. - Cartridges shall be temperature conditioned at -65, +70, and +140 degrees F, for a minimum of four hours and tested for the following phases:

3.7.2 Recoil momentum.- The cartridges shall function satisfactorily and each of the above temperature conditioned samples shall produce a mean launcher recoil momentum of 2.95 ± 0.20 pound-seconds and a standard deviation (σ) not exceeding 0.20 pound-seconds.

3.7.3 Target pattern.- The cartridge when fired shall not fail to release pellets prior to target impact and shall produce a target pattern at 40 meters containing no less than 13 pellets within a 5 foot diameter.

3.7.4 Extraction.- After firing, cartridge cases shall be capable of being extracted from gun chamber manually.

3.7.5 Base plug setback.- After firing, base plugs shall not be more than .010 above flush with case. -65 and +70 degrees F phases need not be fired in this test.

3.7.6 Velocity.- Velocity shall be recorded for informational purposes only.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, refererced specifications and the following:

3.8.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling theitem.

3.8.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

110 cartridges, complete

Ninety of the above cartridges shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured in the same manner using the same materials, equipment, process and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the remaining sample of twenty cartridges shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Lot formation.- A lot shall consist of loaded cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Each lot shall contain:

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4.3.1.1 (Cont'd)

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectiles from one interfix lot number from one manufacturer.

c. Primer from one interfix lot number from one manufacturer.

d. Propellant from one interfix lot number from one manufacturer.

4.3.1.2 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

4.3.2.1 Tool control dimensions.- Dimensions marked tool control shall be gaged at the beginning of each eight hours production from each machine in operation and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted, provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

4.3.2.2 Disc, Drawing A10542392.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Thickness	Gage
MINOR	
201. Burr	Visual
202. Foreign matter	Visual

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4.3.2.3 Plug, Base, Drawing C10542393, covering a detail of Drawing D10542398 (prior to loading propellant and crimping of disc and after insertion of primer).

Defect	Method of Inspection
CRITICAL	
1. Primer anvil missing	Visual
2. Inverted or sidewise primer anvil	Visual
MAJOR	
101. Metal defective	Visual
MINOR	
None defined	

4.3.2.4 Plug, Base, Drawing C10542393, covering a detail of Drawing D10542398 (after loading propellant and crimping of disc).

CRITICAL	
1. Multiple discs	See 6.1.1
MAJOR	
101. Propellant leakage	Visual (see 4.4.4)
MINOR	
None defined	

4.3.2.5 Cartridge 40MM, XM576E1, Drawing D10542398.

CRITICAL	
1. Primer above flush	Gage
MAJOR	
101. Base plug above or below flush of case head	Gage
102. Improper assembly	Visual
103. Sealing compound missing or inadequate (primer)	Visual
104. Sealing compound missing or inadequate (base plug crimp)	Visual
105. Adhesive sealant missing or inadequate (case mouth)	Visual
106. Evidence of primer deformation	Visual
107. Material defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Length, total max.	Gage
202. Marking missing, incorrect or illegible	Visual
203. Burr	Visual
204. Foreign matter	Visual

4.3.2.6 Bandoleer, Drawing F8835107 (before packing in carton).

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Improper number of cartridges	Visual & Manual
202. Plastic support missing	Visual & Manual
203. Bandoleer snaps not closed	Visual

4.3.2.7 Carton, Drawings C8835105 and FA32822 (before sealing).

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Improper number of bandoleers in carton	Visual
202. Separator missing	Visual

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4.3.2.8 Carton, Drawings C8835105 and FA32822 (prior to sealing bag).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Container damaged	Visual
202. Corners not blunted	Visual
203. Contents loose	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.2.9 Wooden packing box, Drawings C8835104 and FA32821 (before sealing).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Marking of bag missing, incorrect or illegible	Visual
202. Bag torn or perforated	Visual
203. Bag improperly sealed	Visual

4.3.2.10 Wooden packing box, Drawings C8835104 and FA32821 (after sealing).

CRITICAL	
None defined	
MAJOR	
None defined	

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Box damaged	Visual
202. Contents loose or broken	Visual
203. Car seal missing or improperly positioned	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Maximum profile gage.- Each cartridge shall be gaged in a Government certified maximum profile gage. Failure of the cartridge to comply with 3.6 shall be cause for rejection of the cartridge.

4.3.4 Primer below flush.- Each cartridge shall be gaged for primer below flush, max. Failure of the cartridge to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.5 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.5.1 Check test for deterioration of primers (See MIL-P-20444).- If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two (2) years, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-P-20444. In addition, primers may be retested for suspicion if mutually agreed upon by Government inspection and the contractor. These tests shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into cartridges (See 6.5).

4.3.5.2 Propellant weight.- When the weight has been established, ten propellant charges after loading into base plug shall be randomly selected from each hour's production or fraction thereof from each loading machine in operation and shall be weighed. Failure of any propellant charge loaded into base plug to comply with 3.4.2 shall be cause for rejection of the production represented by the sample.

4.3.5.3 Primer Misfire Test.- Five base plug-case assemblies from each two hour's production or fraction thereof will be subjected to this test. Failure of any assembly to comply with any of the requirements of 3.4.7 shall be cause for rejection of the production represented by the sample.

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4.3.5.4 Security of projectile crimp.- Five cartridges from each two hour's production or fraction thereof from each crimping machine in operation shall be subjected to this test. Failure of any cartridge to comply with the requirements of the applicable drawing shall be cause for rejection of the production represented by the sample.

4.3.5.5 Waterproof test.- Five cartridges from each four hour's production or fraction thereof shall be ballistically tested to determine compliance with 3.5. Failure of any cartridge to comply shall be cause for rejection of production represented by the sample.

4.3.5.6 Heat seal test of packing seals.- The sample size and acceptance criteria shall be as specified in the applicable specification.

4.3.5.7 Primer Functioning.- Sampling shall be in accordance with paragraph 4.2.3.2.2 of MIL-P-20444B, except that the sample size shall be eight hundred (800) primers from each lot. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected.

4.3.6 Functional.- The following proving ground tests will be performed by the Government.

4.3.6.1 First article sample.

4.3.6.1.1 Initial production sample.- The quantities listed in Table I will be ballistically tested at temperatures indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III, except no retest will be permitted.

TABLE I

Temperature	Quantity
-65 degrees F. ± 5 degrees F.	30
+70 degrees F. ± 5 degrees F.	30
+140 degrees F. ± 5 degrees F.	30

4.3.6.2 Production lot sample.- The quantities listed in Table II will be ballistically tested at temperatures indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III.

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TABLE II

Temperature	Quantity
-65 degrees F. ± 5 degrees F.	20
+70 degrees F. ± 5 degrees F.	20
+140 degrees F. ± 5 degrees F.	20

TABLE III

Defective Characteristic	Classification	Accept	Reject
Primer remains in pocket but is physically loose	Major	10% or less	Over 10%
Failure to extract by spring operation of extractor	Major	10% or less	Over 10%
Failure to extract by hand operation of extractor	Critical	0	1
Target Pattern (see 6.4)	Major	2	3
Primer falls out of pocket	Major	0	1
Blown primer	Major	0	1
Recoil momentum 1/	Major		
Velocity (information only)			
Misfire	Major	0	1

1/ The lot shall be rejected if the mean recoil momentum or standard deviation fails to comply with 3.7.2.

4.3.6.2.1 Suitability assurance test.- Beginning with the first lot produced by each producer and continuing until that producer has submitted three consecutive lots that are accepted in accordance with Table III, an additional sample of ninety cartridges shall be selected from each lot and ballistically tested at temperatures indicated in Table I and the results of the test will be evaluated in accordance with the criteria set forth in Table III. (For information only).

4.3.7 Inspection equipment.- Index of Inspection Equipment List No. IEL-10542398 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format).

4.4 Test methods and procedures.

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4.4.1 Propellant weight.- The samples provided in accordance with 4.3.5.2 shall be weighed by using an approved scale. The scale shall be calibrated prior to every four hour's production. If check weighing is to be performed, check weighing shall be accomplished independently of the production weighing system using a different scale.

4.4.2 Security of projectile crimp.- The samples supplied in accordance with 4.3.5.2 shall be mounted in an approved pull test device in such a manner that the cartridge case will be held stationary. The force will be applied to the sabot until it is separated from the case. Force required to achieve separation must be within specified limits.

4.4.3 Primer Misfire Test.- The samples supplied in accordance with 4.3.5.3 shall be temperature conditioned at +70 degrees F. and fired from an approved launcher.

4.4.4 Propellant leakage.- The samples supplied in accordance with 4.3.2.4 shall be vigorously shaken a minimum of six times with the disc positioned in a downward direction.

4.4.5 Waterproof test.- The samples supplied in accordance with 4.3.5.4 shall be temperature conditioned at +70 degrees F. and placed base down on a screen approximately one-half inch from the bottom of a container with a six inch head of water (+70 degrees F) for twelve hours, after which the rounds will be dried and fired within four hours.

4.4.6 Heat seat test of packing seals.- The procedures for this test shall be in accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

4.4.7 Functional

4.4.7.1 Ballistic test.- The cartridges supplied in accordance with 4.3.6.1 or 4.3.6.2 will be fired from an M-79 launcher mounted on a recoil measuring pendulum. The gun is fired by means of a solenoid. Firing the gun causes the pendulum to recoil and by means of a stylus attached to the pendulum, the length of swing is marked on a flat surface beneath it. This recoil distance is converted to momentum (lb-sec) using total weight of weapon and pendulum and the period of the pendulum (time required for pendulum to swing through one complete cycle) (see 6.3). Velocity shall be measured at 7.5 feet from the muzzle of the launcher using break screens placed 5 feet and 10 feet from the muzzle. A target measuring 8 feet by 8 feet shall be placed 40 meters from the muzzle of the launcher. Observations shall be made for the defective characteristics listed in Table III and recorded for each round ballistically tested. (Ref. TECP 700-700, Interim Pamphlet 20-20).

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4.4.7.1.1 Suitability assurance test.- From the samples selected in accordance with 4.3.6.2.1 the cartridges shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, test number 104. After transportation vibration, the cartridges shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, test number 105. Following the temperature and humidity cycle, the cartridges shall be temperature conditioned in accordance with Table I. The additional sample of ninety cartridges will be fired as specified in 4.4.7.1.

4.4.8 Primer Functioning.- Each primer of the 800 samples shall be tested as specified in MIL-P-20444B, paragraph 4.3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A (world wide shipment).- The cartridges shall be packed in accordance with Drawings C8835104, FA32821 and FA32822.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

5.3 Data card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.1.1 Contractor may use any control method (i.e., gage, measure, visual, etc.) to assure that no more than one (1) disc is assembled to each base plug. The control procedure and inspection plan shall be forwarded to the responsible Technical Engineering Agency for approval prior to its implementation into production.

6.2 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.3 Converting to momentum (lb-sec).- The formula used to convert the distance measurement to lb-sec is as follows:

$$mv = \frac{\pi wd}{6gT}$$

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where w = weight of pendulum
d = distance of recoil in inches
g = gravity in ft/sec^2
T = period of pendulum swing

The period of the pendulum swing is determined by displacing the pendulum rearward about five or six inches and then allowing it to swing freely. Using a stopwatch, measure the time it takes for the pendulum to swing through ten cycles. Repeat this procedure three times and take an average of the measurements as the value of T. The period of the pendulum swing should be checked as a calibration procedure before each new lot of cartridges is fired for acceptance.

6.4 Target pattern at -65 degrees F. ± 5 degrees F., is for information only.

6.5 Cost of Check Test.- The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

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